

Li_3YCl_6 — crystalline transport benchmark

From experimental reference to a controlled three-model MD comparison

1 Literature

Asano et al. (2018)
Experiment: chloride SSE
RT conductivity $>1 \text{ mS cm}^{-1}$

2 Model & software

Explicit Li/Y ordering
 $2 \times 2 \times 2$; 240 atoms
MACE / SevenNet / M3GNet
LAMMPS; 1 GPU per run

3 MD conditions

T: 400 / 600 / 800 / 1000 K
Ensemble: fixed-cell NVT
dt: 1 fs
Equil.: 50 ps
Production: 500 ps

4 Analysis

Li MSD $\rightarrow D(T)$
Arrhenius fit $\rightarrow E_a$
300 K extrapolation
Conditional σ_{NE}

5 Conclusion

Thermal trend reproduced
Absolute transport is
strongly model dependent